

Applicants Dataset Insights

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1. INTRODUCTION

This report presents an Exploratory Data Analysis (EDA) of the applicants dataset for my data science internship at NextGenLearners. The objective is to identify patterns in applicant domains, application trends, application status, and university-wise participation using data visualization techniques. The insights obtained from this analysis can help improve future recruitment and outreach strategies.

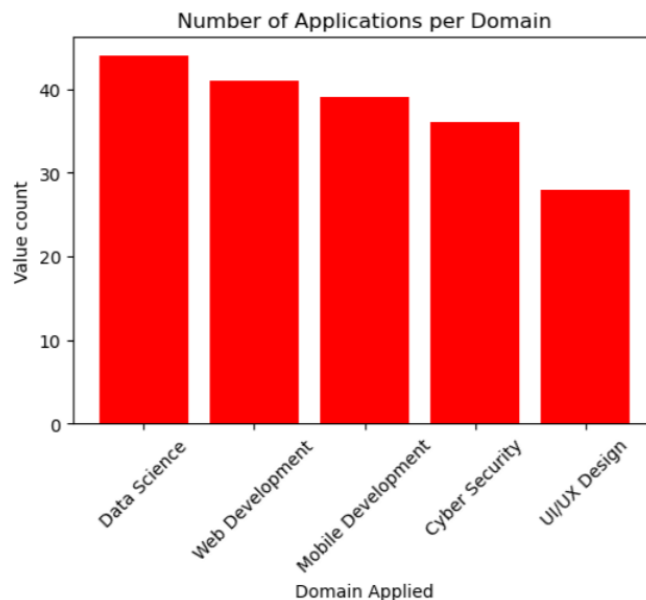
2. EXPLORATORY DATA ANALYSIS

2.1 Number of Applications by Domain

Purpose:

To identify which domains received the highest number of applications.

Chart:



Key Insights:

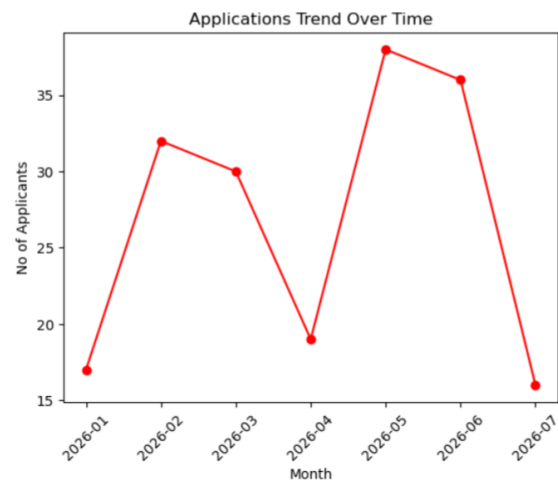
- Data Science received the highest number of applications, making it the most preferred domain among applicants.
- Web Development ranked second, showing strong interest in software development careers.
- Mobile Development also attracted a considerable number of applications, indicating steady demand in mobile technologies.
- Cyber Security received a moderate number of applications, reflecting growing awareness of security-related careers.
- UI/UX Design recorded the lowest number of applications, suggesting comparatively lower applicant interest.
- Overall, technical and programming-focused domains attracted more applicants than design-oriented fields.

2.2 Applications Trend Over Time

Purpose

To analyze how application submissions changed over time.

Chart:



Key Insights:

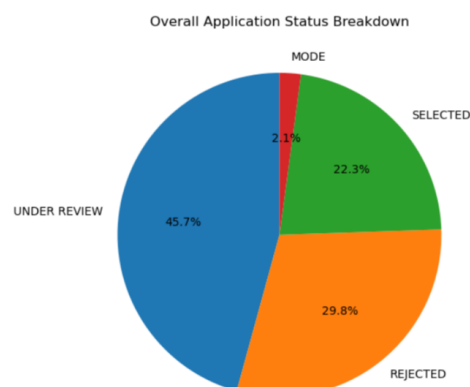
- Applications increased significantly from January to February, indicating an early rise in applicant interest.
- A slight decline was observed during March and April, suggesting a temporary slowdown in application submissions.
- May recorded the highest number of applications, making it the peak month of the recruitment period.
- Application numbers remained relatively high in June, showing continued engagement from applicants.
- A sharp decline occurred in July, indicating that most applicants had already submitted their applications before the final month.
- The trend suggests that promotional activities were most effective during the middle of the recruitment period.

2.3 Overall Application Status Breakdown

Purpose

To understand the overall distribution of application statuses.

Chart:



Key Insights:

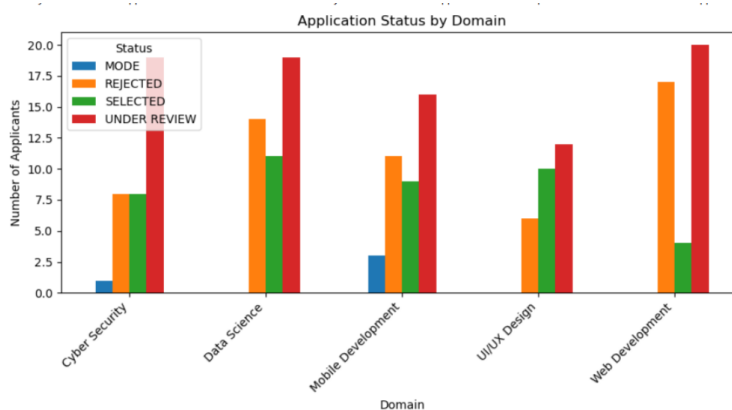
- Nearly 46% of applications are currently Under Review, making it the largest status category.
- Approximately 30% of applicants were Rejected, indicating a competitive selection process.
- Around 22% of applicants have been Selected so far.
- The high percentage of applications under review suggests that the recruitment process is still ongoing.
- Selection decisions may change once all pending applications have been evaluated.

2.4 Application Status by Domain

Purpose

To compare application statuses across different domains.

Chart:



Key Insights:

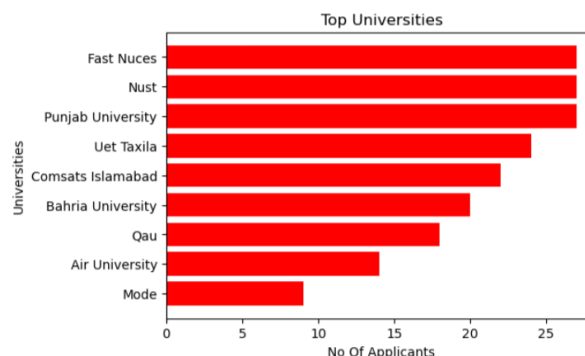
- Web Development has the highest number of applications currently under review.
- Data Science recorded the highest number of selected applicants among all domains.
- Cyber Security also has a large number of applications awaiting review.
- UI/UX Design has the lowest number of applicants across almost all status categories.
- Rejected applications are distributed across all domains, with Web Development and Data Science showing relatively higher counts.

2.5 Top Universities by Number of Applicants

Purpose

To identify the universities contributing the most applicants.

Chart:



Key Insights:

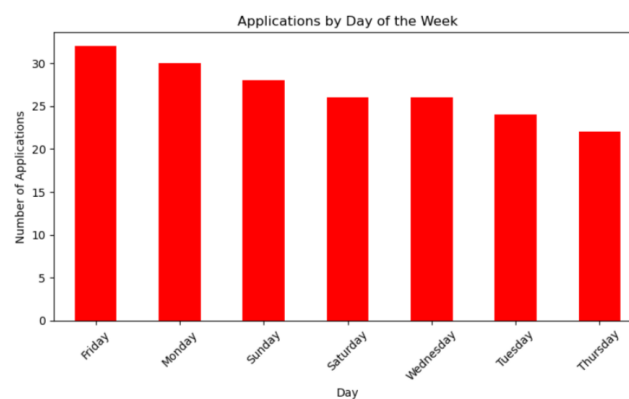
- FAST NUCES, NUST, and Punjab University received the highest number of applications.
- UET Taxila and COMSATS Islamabad also contributed a significant number of applicants.
- Most applications originated from a small group of leading universities.
- Universities with strong technology and engineering programs contributed the majority of applicants.
- Air University, QAU, and Bahria University had comparatively fewer applications.
- These findings can help focus future recruitment and awareness campaigns on both high- and low-participation universities.

2.6 Applications by Day of the Week

Purpose

To determine whether applications are submitted more frequently on particular days of the week.

Chart:



Key Insights:

- Friday recorded the highest number of application submissions.
- Monday also showed a high level of applicant activity.
- Application submissions remained fairly consistent during the rest of the week.
- Thursday had the lowest number of applications.
- Applicants appeared to be more active at the beginning and end of the academic week.
- These patterns can help schedule reminders and promotional announcements on days with higher applicant engagement.

SHORT SUMMARY:

- Data Science attracted the highest number of applicants, indicating strong interest in this domain.
- Applications peaked during May and June, suggesting this period was the most effective for attracting applicants.
- Nearly half of all applications are still under review, while approximately one-fifth have been selected.
- FAST NUCES, NUST, and Punjab University contributed the largest number of applicants, making them key institutions for future outreach.
- Application submissions were highest on Friday, suggesting applicants were more active toward the end of the academic week.